

ModelGuard

**Blockchain-based AI model provenance
and integrity verification**

Group 11

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AI models are files, and files can be swapped

Replaced

An attacker swaps the official file on a mirror or download server.

Backdoored

A few weights are changed so the model misbehaves on chosen inputs.

Mislabeledled

An unofficial file is presented as an official release or version.

Still in use

A version the publisher withdrew keeps running in production.



One changed byte, a completely different fingerprint

SHA-256 turns any file into a 32-byte fingerprint. We flipped 4 bits in our demo model, mnist-12.onnx (26 KB):

Original

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

Tampered (4 bits flipped)

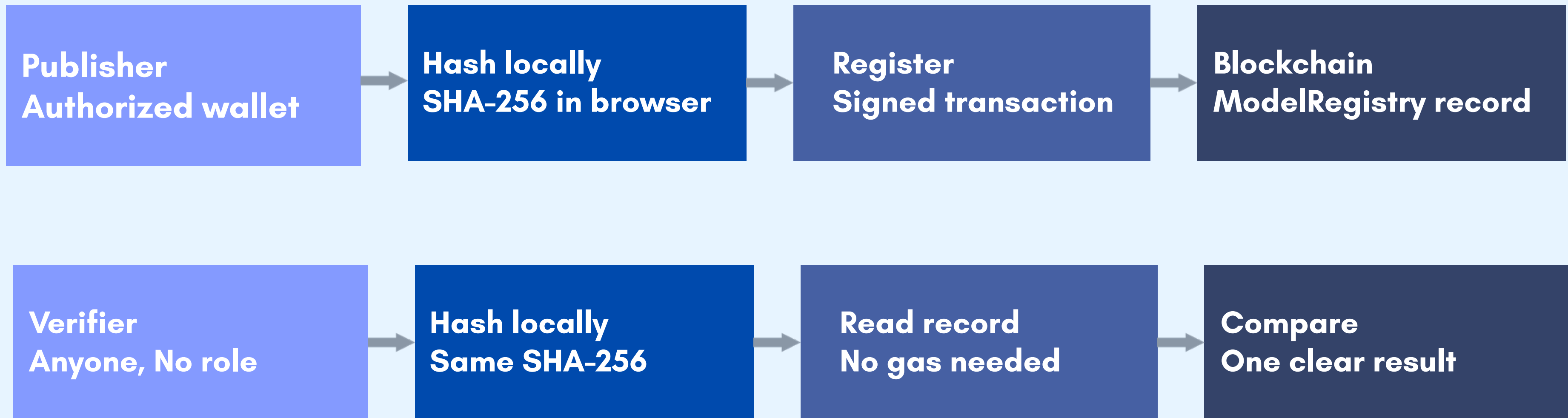
0x3b18a736ad8af386be07eb9df87b5d75e659efccc4fc178afd52b4edfd648f8f

ModelGuard puts the original fingerprint on a blockchain, where no one can quietly rewrite it.

Technology Stack

Layer	Technology	Why
Smart contract	Solidity 0.8, OpenZeppelin	Audited role-based access control
Tooling	Hardhat 3, TypeScript	Compile, test, deploy with Ignition
Blockchain	Local Hardhat network	Free, fast, repeatable demo
Frontend	React with Vite	Publisher and verifier screens
Chain client	ethers v6	Wallet and contract calls
Wallet	MetaMask	Identifies publishers, signs transactions
Hashing	Web Crypto SHA-256	Hash in browser, no upload

How ModelGuard works



Model files never leave the user's computer. Only 32-byte hashes and small metadata go on-chain.

Five steps, one model

Wallet	Action	Expected Result
Publisher	Register DemoClassifier 1.0.0	Record created
Ordinary verifier	Verify the original file	Verified
Ordinary verifier	Revoke 1.0.0, then re-verify	Hash mismatch
Publisher	Revoke 1.0.0, then re-verify	Revoked warning
Publisher	Register version 1.1.0	Both versions visible

What the verifier sees

PUBLISHER ACTION

Register a model

Create a permanent blockchain record for a new model version. Files are hashed in your browser and are never uploaded.

Model name

Case-sensitive, e.g. DemoClassifier

DemoClassifier

Version

Semantic version, e.g. 1.0.0

1.0.0

Model file

The exact artifact to publish (ONNX, TFLite, PyTorch...)

Choose File

mnist-12-original.onnx

Provenance manifest

JSON file describing source, licence, and dataset

Choose File

mnist-12-manifest.json

Metadata URI (optional)

Public link to release notes or the manifest

https://... or ipfs://...

Hash files and register

Clear

MODEL SHA-256

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

PROVENANCE MANIFEST SHA-256

0xa8dea1e117fdaca553263b0862f7f17b8230d98e319e4ea98ca026ce81dc6d6d

✔ Prepare

✔ Confirm in MetaMask

✔ Wait for block

✔ Confirmed

DemoClassifier 1.0.0 is registered. Anyone can now verify it in the Verify panel.

Transaction 0x3bd28a04cb5926fae9977e2770303fdc954cc682759d2b337eb6b8db2116d062 · block 2

Registration

PUBLIC ACTION

Verify a model

Check that a downloaded model is byte-for-byte identical to the registered release. No transaction or gas

Model name

Exactly as registered

DemoClassifier

Version

e.g. 1.0.0

1.0.0

Model file to check

Hashed locally with SHA-256; the file never leaves your device

Choose File

mnist-12-original.onnx

Verify file

Clear

✔ Verified — active registered model

This file is byte-for-byte identical to DemoClassifier 1.0.0, registered by 0xf39F...2266 on 9/29/2026, 1:2

YOUR FILE (SHA-256)

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

REGISTERED ON-CHAIN

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

✔ Fingerprints match

Verification

PUBLIC ACTION

Verify a model

Check that a downloaded model is byte-for-byte identical to the registered release. No transaction or gas is n

Model name

Exactly as registered

Version

e.g. 1.0.0

DemoClassifier

1.0.0

Model file to check

Hashed locally with SHA-256; the file never leaves your device

Choose File

mnist-12-modified.onnx

Verify file

Clear



Hash mismatch — file may have been modified

This file is not the registered DemoClassifier 1.0.0. Even a one-byte change produces a different finger

YOUR FILE (SHA-256)

0x3b18a736ad8af386be07eb9df87b5d75e659efccc4fc178afd52b4edfd648f8f

REGISTERED ON-CHAIN

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

✗ Fingerprints differ

Hash mismatch

PUBLIC ACTION

Verify a model

Check that a downloaded model is byte-for-byte identical to the registered release. No transaction or gas is n

Model name

Exactly as registered

Version

e.g. 1.0.0

DemoClassifier

1.0.0

Model file to check

Hashed locally with SHA-256; the file never leaves your device

Choose File

mnist-12-original.onnx

Verify file

Clear



Revoked — this version must not be used

The file matches the registered DemoClassifier 1.0.0, but the publisher has revoked this version.

Reason given: Weights found to be compromised

YOUR FILE (SHA-256)

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

REGISTERED ON-CHAIN

0x5c688690f8bacf667d4c2074af5ad0646ca328d7ab03eccf944a65b320171bdd

✓ Fingerprints match

Revoked



What the blockchain adds

- 01. Tamper detection**
Any change to the file, even one bit, produces a mismatch.
- 02. Authorized publishing**
Only wallets with the publisher role can register; the contract enforces it.
- 03. Permanent history**
Records can't be overwritten or deleted, and duplicates are rejected.
- 04. Visible revocation**
Withdrawn versions stay on record with a public reason.

Thank you!

